



AQI_predict-1.pdf

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Page 1 / 8



100%



1

Pearls AQI Predictor

Let's predict the Air Quality Index (AQI) in your city in the next 3 days, using a 100% serverless stack.

2

PROJECT

The following is a high level overview for you to achieve this:



3

Feature pipeline

Write a Python script that:

- 1 - Fetches raw sensor and published data from an external API like AQICN or OpenWeatherMap.
- 2 - Computes features from this raw data (date, model inputs), and targets (like model outputs).
- 3 - Stores features in the Feature Store.



4

Backfill historical (features, targets)

Run the Feature script from step 1 for a range of past dates, to generate training data for your ML models.



5

Training pipeline

Write a Python script that:

- 1 - Fetches historical (features, targets) from the Feature Store.
- 2 - Trains and evaluates the ML model (using the training data) for this data.
- 3 - Stores the trained model in the Model Registry.



6

Automate pipeline runs

Create a CI/CD pipeline that automatically runs:

- the Feature script every hour, and
- the training script every day.

Remember any CI/CD tool that you can use on Kubernetes and GitHub Actions for you are encouraged to explore other tools too.



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